

PROGRAM-5 Demonstrate CI integration by running jenkins in a Docker container

Program structure:

Dockerfile

Docker-compose.yml

Step 1: check where docker service is running

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog5$ docker version
```

```
Client: Docker Engine - Community
```

```
Version:      28.5.1
API version:  1.51
Go version:   go1.24.8
Git commit:   e180ab8
Built:        Wed Oct  8 12:17:26 2025
OS/Arch:      linux/amd64
Context:      default
```

```
Server: Docker Engine - Community
```

```
Engine:
Version:      28.5.1
API version:  1.51 (minimum version 1.24)
Go version:   go1.24.8
Git commit:   f8215cc
Built:        Wed Oct  8 12:17:26 2025
OS/Arch:      linux/amd64
Experimental: false
containerd:
Version:      v1.7.28
GitCommit:    b98a3aace656320842a23f4a392a33f46af97866
runc:
Version:      1.3.0
GitCommit:    v1.3.0-0-g4ca628d1
docker-init:
Version:      0.19.0
GitCommit:    de40ad0
```

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog5$ docker compose version
```

```
Docker Compose version v2.40.0
```

Step2: Create a Dockerfile

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog5$ cat Dockerfile
```

```
FROM jenkins/jenkins:lts
```

```
USER root
```

```
RUN apt-get update && apt-get install -y docker.io
```

```
USER jenkins
```

Step3: Create a docker-compose.yml

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog5$ cat docker-compose.yml
version: '3.8'
```

```
services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
    # optional: allow container processes to be in host docker group (recommended)
    # set DOCKER_GID in your shell before running `docker compose up`
    group_add:
      - "${DOCKER_GID:-999}"
    restart: unless-stopped

volumes:
  jenkins_home:
```

Step-4: Build the docker image

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog5$ sudo docker build -t prog5 .
[sudo] password for 1RV24MC025_CHANDAN_K:
[+] Building 72.8s (7/7) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 136B                               0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts   3.7s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e487df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 31.0s
=> resolve docker.io/jenkins/jenkins:lts@sha256:1e487df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 0.0s
=> sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa812cc605644 12.71kB / 12.71kB 0.0s
=> sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 65.08MB / 65.08MB 19.7s
=> sha256:1e487df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 3.12kB / 3.12kB 0.0s
=> sha256:68cde4f8009293f6a12f5c50b91f9cf4194bd6293334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB 1.5s
=> sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befd8e03923e093b000 2.58kB / 2.58kB 0.0s
=> sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB 2.2s
=> sha256:e0e73f34dee9332ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B 2.7s
=> sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d0688c35e7f1ed8627 95.32MB / 95.32MB 29.7s
=> sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaa0b98 189B / 189B 3.7s
=> sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2f8f88e9c009f6874e7221acb06245 6.29MB / 6.29MB 5.9s
=> sha256:d7dd2e63023897146ec4b913a460382968fbdd1cc3682e36f650727a1bf6d6899 72.05MB / 72.05MB 28.5s
=> extracting sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 0.9s
=> sha256:3fb70446889ebcc58e91f43f90aa314798941c318ce016ea62068a3de53927 2.23kB / 2.23kB 20.8s
=> extracting sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 0.1s
=> extracting sha256:68cde4f8009293f6a12f5c50b91f9cf4194bd6293334d8a31a45b1ed63cb91a5 0.0s
=> sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8bbeea 1.29kB / 1.29kB 21.9s
=> extracting sha256:e0e73f34dee9332ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 0.0s
=> sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313 392B / 392B 23.0s
=> extracting sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d0688c35e7f1ed8627 0.2s
=> extracting sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaa0b98 0.0s
=> extracting sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2f8f88e9c009f6874e7221acb06245 0.0s
=> extracting sha256:d7dd2e63023897146ec4b913a460382968fbdd1cc3682e36f650727a1bf6d6899 0.2s
=> extracting sha256:3fb70446889ebcc58e91f43f90aa314798941c318ce016ea62068a3de53927 0.0s
=> extracting sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8bbeea 0.0s
=> extracting sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313 0.0s
[2/2] RUN apt-get update && apt-get install -y docker.io      37.0s
=> exporting to image                                           0.9s
=> exporting layers                                              0.8s
=> writing image sha256:916e707501a02d2e8ae6cebd84b1259fc2441051e278304c86838e470bbd0693 0.0s
=> naming to docker.io/library/prog5                            0.0s
```

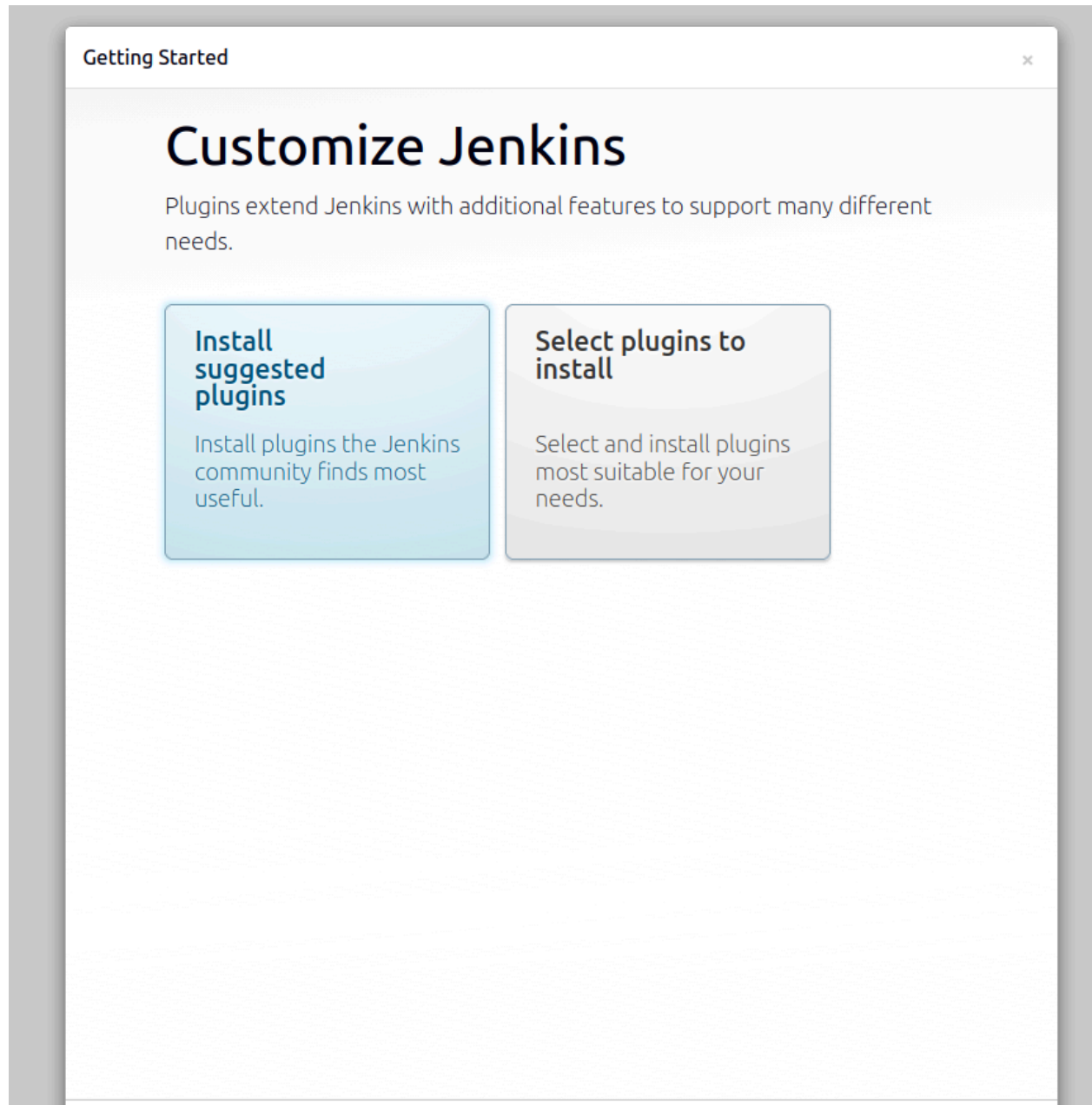
Step-5: Run the docker image

```

1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog5$ sudo docker run prog5
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-01-07 10:28:36.571+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-01-07 10:28:37.107+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-01-07 10:28:37.139+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea;
jvm 21.0.9+10-LTS
2026-01-07 10:28:37.376+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-01-07 10:28:37.411+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-01-07 10:28:37.646+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2026-01-07 10:28:37.713+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3c782d8e[Jenkins v2.528.3/,b=file:///var/jenki
ns_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9[STARTED]]
2026-01-07 10:28:37.723+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@5c60b0a0[HTTP/1.1, (http/1.1)][0.0.0.0:8080]
2026-01-07 10:28:37.731+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@3668d4[STARTING][12.0.25,sto=0] @1542ms
2026-01-07 10:28:37.731+0000 [id=55] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-01-07 10:28:37.813+0000 [id=53] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.3
2026-01-07 10:28:37.877+0000 [id=63] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-01-07 10:28:37.885+0000 [id=83] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-01-07 10:28:38.354+0000 [id=92] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-01-07 10:28:38.358+0000 [id=64] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-01-07 10:28:38.359+0000 [id=64] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-01-07 10:28:38.453+0000 [id=86] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-01-07 10:28:38.453+0000 [id=88] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-01-07 10:28:38.454+0000 [id=90] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-01-07 10:28:38.455+0000 [id=90] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-01-07 10:28:38.479+0000 [id=106] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-01-07 10:28:38.711+0000 [id=74] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 7476885f0dc3490597a6f0bbfd6dd686
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****

```

Step-6: Verify whether jenkins is running by navigation to <http://localhost:8080>



Step-7: Display the docker images and remove the image

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog5$ sudo docker rmi -f prog5-jenkins
Untagged: prog5-jenkins:latest
```